

Complete Splunk SIEM Lab Implementation Guide

Windows 11 → Splunk Universal Forwarder → Kali Linux Splunk Enterprise

Objective

Build a complete SOC/SIEM lab where Windows 11 forwards logs to Splunk Enterprise running on Kali Linux for real-time monitoring and analysis.

Architecture Overview

Windows 11 generates logs → Splunk Universal Forwarder sends logs → Kali Linux Splunk Enterprise receives, indexes, and searches logs.

Lab Requirements

1. VMware Workstation
2. Kali Linux VM (4GB RAM recommended)
3. Windows 11 (8GB RAM recommended)
4. NAT or Bridged networking

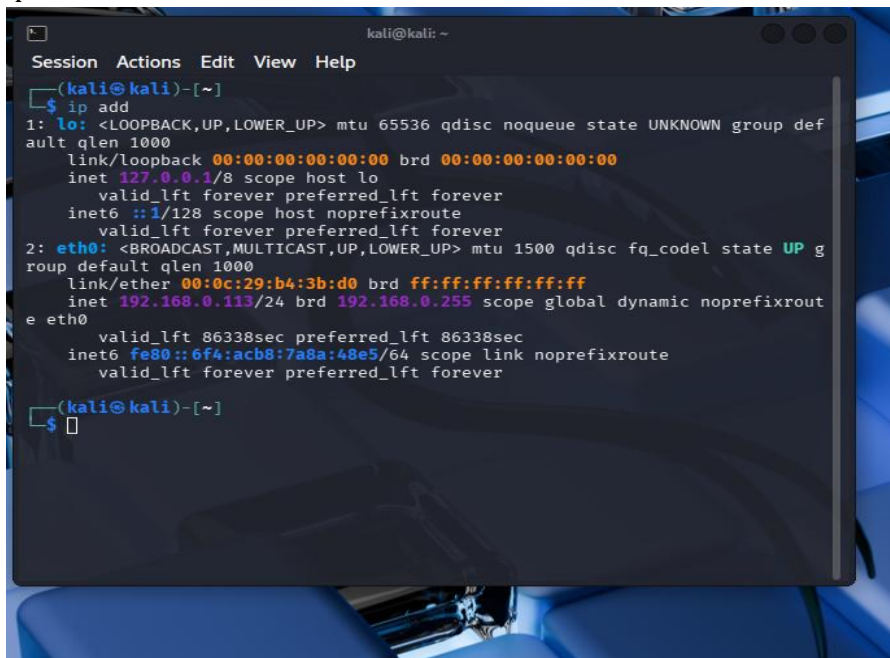
Network Configuration

Use NAT or Bridged networking in VMware so VM can communicate.

Verify Connectivity

On Kali:

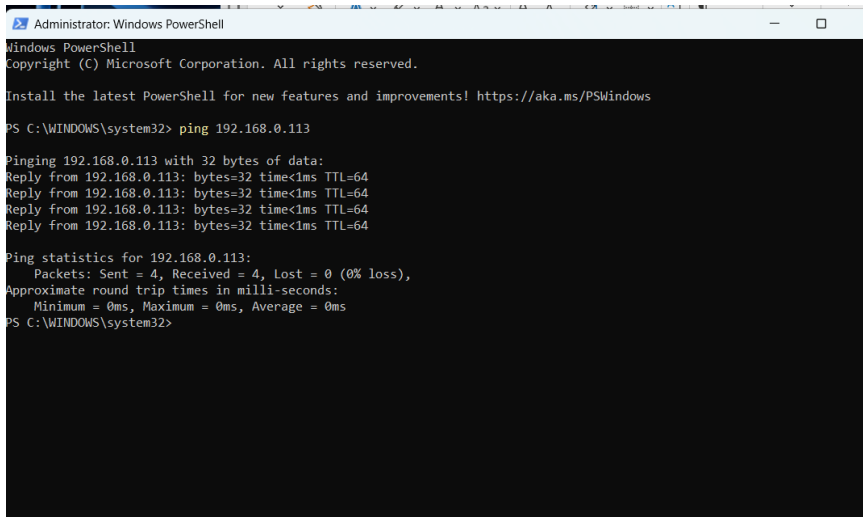
ip add



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ip add  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 00:0c:29:b4:3b:d0 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.0.113/24 brd 192.168.0.255 scope global dynamic noprefixrout  
e eth0  
        valid_lft 86338sec preferred_lft 86338sec  
    inet6 fe80::6f4:acb8:7a8a:48e5/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
(kali@kali)-[~]  
$
```

On Windows:

ping <kali-ip>



```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> ping 192.168.0.113

Pinging 192.168.0.113 with 32 bytes of data:
Reply from 192.168.0.113: bytes=32 time<1ms TTL=64
Reply from 192.168.0.113: bytes=32 time<1ms TTL=64
Reply from 192.168.0.113: bytes=32 time<1ms TTL=64
Reply from 192.168.0.113: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.0.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\WINDOWS\system32>
```

Installing Splunk Enterprise on Kali

Download Splunk:

wget -O splunk.deb

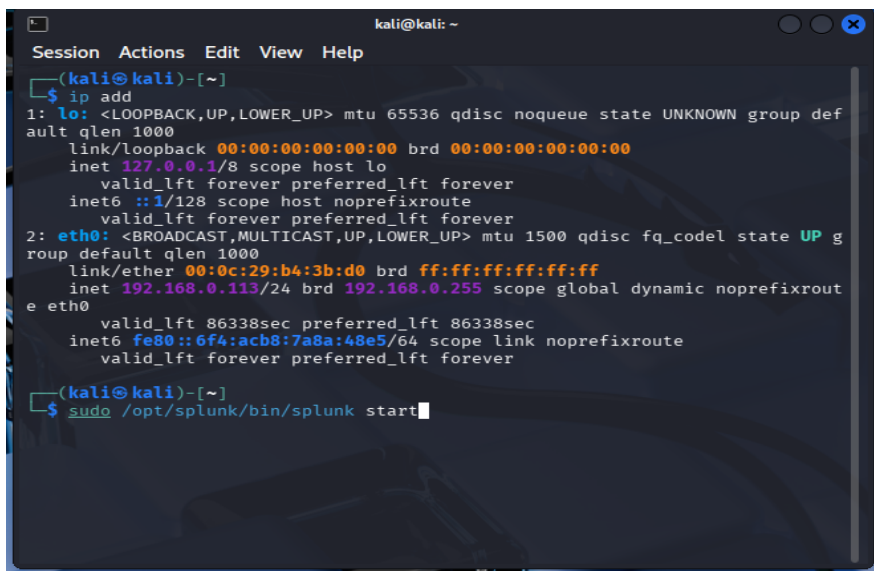
"<https://download.splunk.com/products/splunk/releases/10.0.2/linux/splunk-10.0.2-e2d18b4767e9-linux-amd64.deb>"

Install:

sudo dpkg -i splunk.deb

Start Splunk:

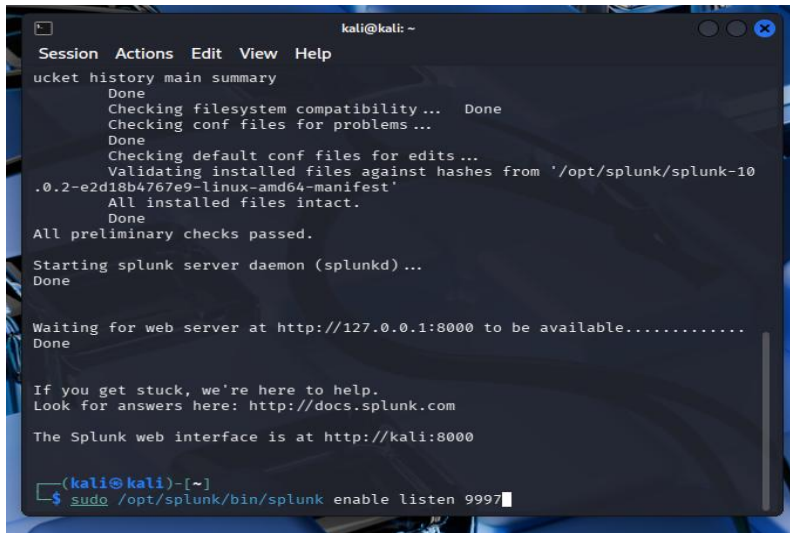
sudo /opt/splunk/bin/splunk start



```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)~$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b4:3b:d0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.113/24 brd 192.168.0.255 scope global dynamic noprefixroute eth0
        valid_lft 86338sec preferred_lft 86338sec
    inet6 fe80::6f4:acb8:7a8a:48e5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(kali@kali)~$ sudo /opt/splunk/bin/splunk start
```

Enable Receiving Port

`sudo /opt/splunk/bin/splunk enable listen 9997`

A screenshot of a terminal window titled 'kali@kali: ~'. The window shows the output of the 'sudo /opt/splunk/bin/splunk enable listen 9997' command. The output includes a progress bar, status messages like 'Checking filesystem compatibility...', 'Checking conf files for problems...', 'Checking default conf files for edits...', 'Validating installed files against hashes from '/opt/splunk/splunk-10.0.2-e2d18b4767e9-linux-amd64-manifest'', 'All installed files intact.', 'All preliminary checks passed.', 'Starting splunk server daemon (splunkd)...', 'Waiting for web server at http://127.0.0.1:8000 to be available...', and a final message: 'If you get stuck, we're here to help. Look for answers here: http://docs.splunk.com. The Splunk web interface is at http://kali:8000'. The terminal prompt is '(kali@kali)-[~]' and the command being entered is 'sudo /opt/splunk/bin/splunk enable listen 9997'.

Verify:

`sudo ss -tunlp | grep 9997`

Installing Splunk Universal Forwarder on Windows 11

PowerShell Download Command:

`Invoke-WebRequest -Uri`

`"https://download.splunk.com/products/universalforwarder/releases/10.0.2/windows/splunkforwarder-10.0.2-x64-release.msi" -OutFile`

`"$env:USERPROFILE\Downloads\splunkforwarder.msi"`

Install using:

`msiexec.exe /i "$env:USERPROFILE\Downloads\splunkforwarder.msi"`

Configure Forwarder in Splunk Universal Forwarder on Windows 11

During installation, set receiving indexer to:

`<KALI-IP>:9997`

Configure Windows Event Logs Splunk Universal Forwarder on Windows 11

Create inputs.conf in:

`C:\Program Files\SplunkUniversalForwarder\etc\system\local`

Add:

`[WinEventLog://Application]`

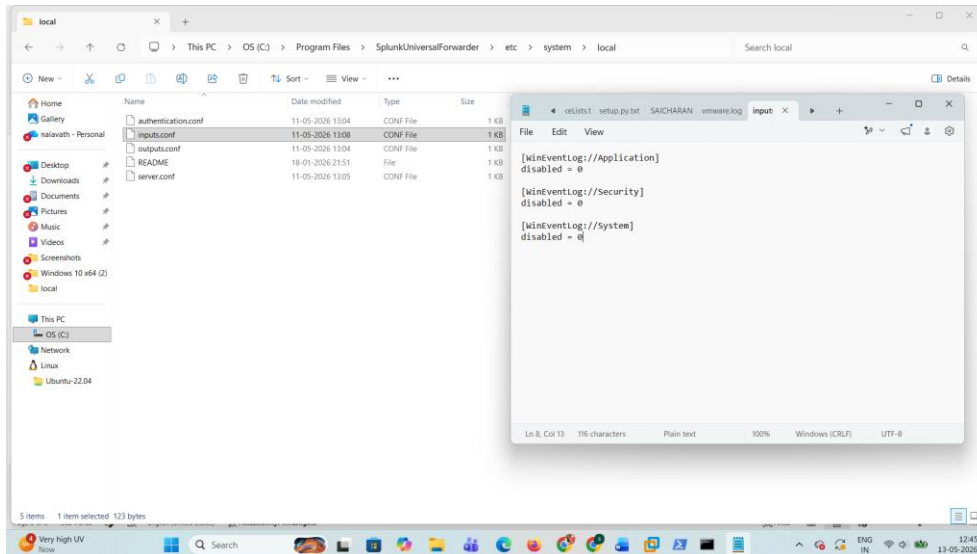
`disabled = 0`

`[WinEventLog://Security]`

disabled = 0

[WinEventLog://System]

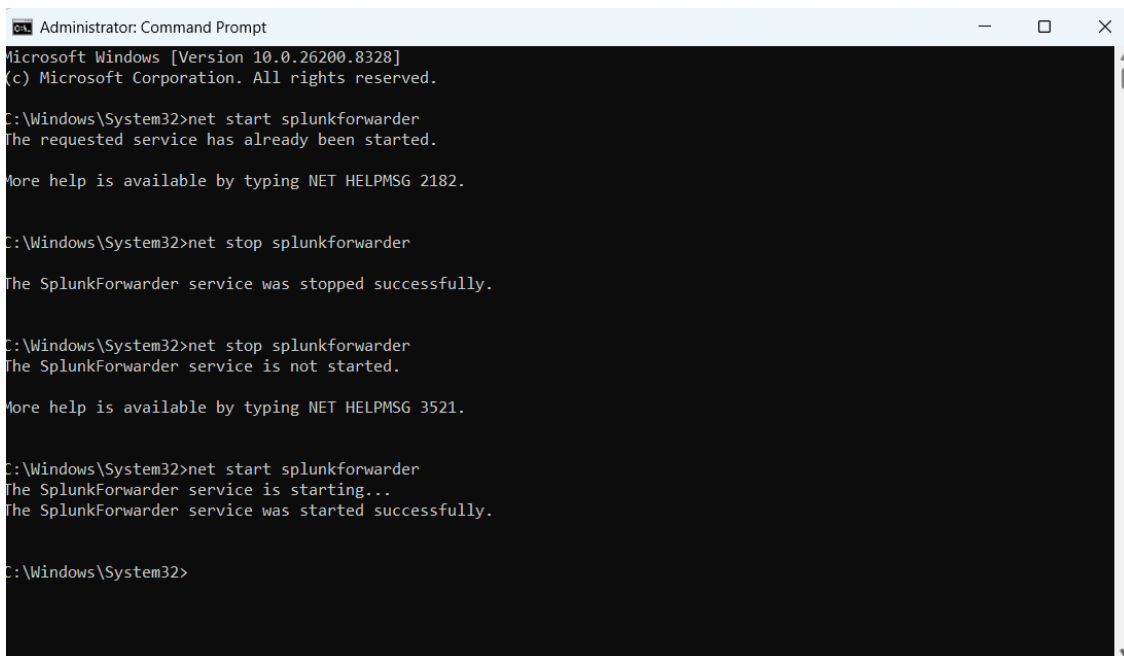
disabled = 0



In the Windows Power shell Restart Forwarder

net stop splunkforwarder

net start splunkforwarder



In the Windows Power shell Verify Connection

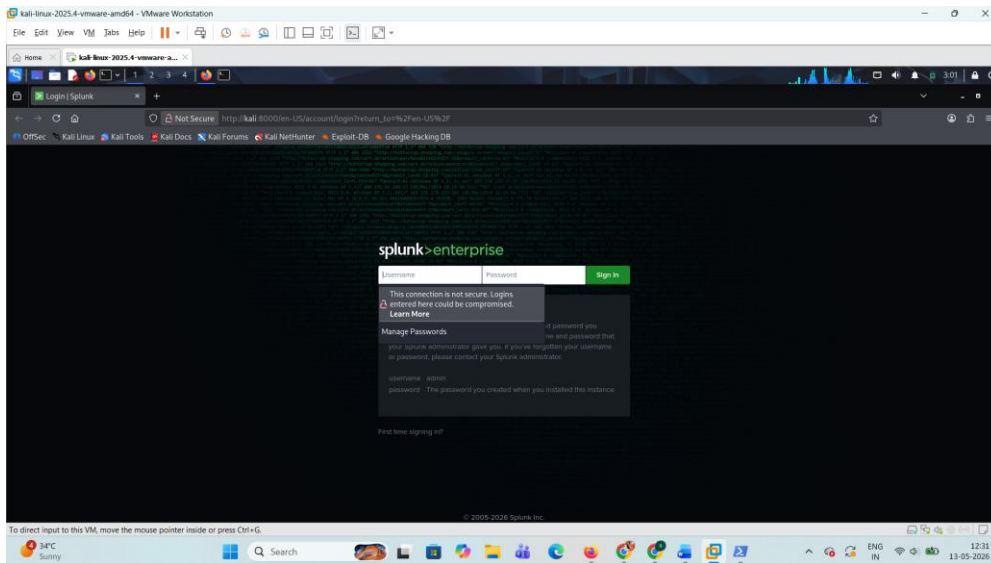
PowerShell:

Test-NetConnection <KALI-IP> -Port 9997

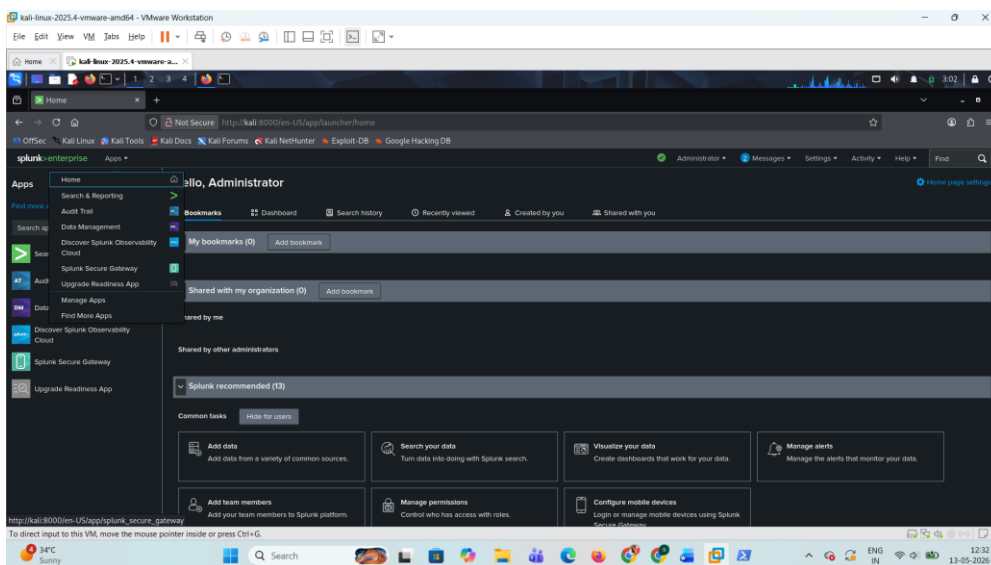
View Logs in Splunk

Open:

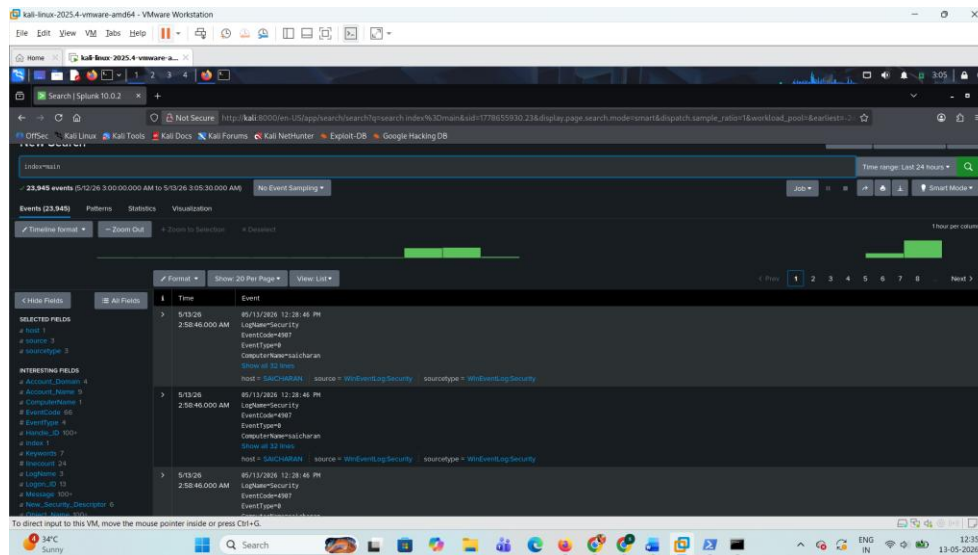
<http://localhost:8000>



Go to Apps → Search & Reporting



Search:
index=*



Useful SPL Queries

All Windows Logs:
`sourcetype="WinEventLog:*"`

Security Logs:
`sourcetype="WinEventLog:Security"`

Failed Login Events:
`EventCode=4625`

Successful Login Events:
`EventCode=4624`

Generate Test Logs

PowerShell Simulation:

```
for ($i=0; $i -lt 20; $i++) {  
    net use \\localhost\C$ /user:FakeUser WrongPassword123  
}
```

Common Troubleshooting

1. No logs → Check inputs.conf
2. Port issue → Enable port 9997
3. No connection → Verify VMware networking
4. Splunk not opening → Restart Splunk

Advanced Improvements

You can later add:

- Sysmon
- Apache/Nginx logs
- Wazuh
- Suricata IDS
- Azure cloud logs

Resume Statement

Built a SIEM lab using Splunk Enterprise on Kali Linux and Splunk Universal Forwarder on Windows 10 to collect, forward, index, and analyze real-time Windows security logs.